Truncated Burgers-Hopf dynamics

$$u_t + \frac{1}{2}\partial_x(u^2) = 0 \tag{1}$$

$$u = \sum_{k=-K}^{K} \hat{u}_k e^{ikx} \text{ with } \hat{u}_{-k} = \hat{u}k^* \tag{2}$$

$$u^2 := w = \sum_{\ell=-K}^{K} \sum_{m=-K}^{K} \hat{u}_\ell \hat{u}_m e^{i(\ell+m)x} \tag{3}$$

$$= \sum_{k=-K}^{K} \hat{w}_k e^{ikx} \tag{4}$$

$$\therefore \hat{w}_k = \sum_{m=-K}^{K} \widehat{u}_m \widehat{u}_{k-m} \tag{5}$$